

LEARNING BY LOOKING BACK: MOMENTUM-BASED TRAJECTORY DEBIASING FOR CONTRASTIVE LEARNING

Anonymous authors

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ABSTRACT

Contrastive self-supervised learning thrives on differentiating an anchor from a large set of negative samples. Its performance, however, is critically undermined by the presence of ‘false negatives’—semantically similar instances that are inadvertently repelled, leading to a fragmented embedding space. This is particularly challenging because effective learning requires ‘hard negatives’, which are often visually indistinguishable from false negatives. Existing methods attempt to solve this by using static class priors or indirect heuristics like score variance, but these approaches are either too coarse or rely on ambiguous signals. A stable, true hard negative can appear identical to a false negative under these heuristics. We propose Momentum-based Trajectory Debiasing (MTS), a novel method that identifies false negatives by leveraging a direct and unambiguous signal: the temporal trajectory of similarity scores. By exploiting the inherent temporal gap between a standard online encoder and its slower-moving momentum counterpart, MTS computes a state-free proxy for the similarity gradient. A consistently increasing similarity is a strong indicator of a false negative. This signal is used to dynamically and selectively reduce the repulsive force on suspected false negatives within the contrastive loss. Experiments on the CIFAR-100 fine-grained benchmark show that MTS significantly outperforms standard contrastive learning and existing debiasing techniques, demonstrating its ability to learn more robust and semantically coherent representations.

1 INTRODUCTION

Self-supervised learning, particularly through contrastive methods, has emerged as a powerful paradigm for learning visual representations without manual labels. Frameworks like MoCo Kalanidis et al. (2020) learn by pulling together representations of augmented views of the same image (positives) while pushing them apart from representations of other images (negatives). A crucial ingredient for success is the use of hard negatives—samples that are difficult to distinguish from the anchor—as they provide the most informative gradient signals for learning fine-grained decision boundaries Robinson et al. (2020). However, this reliance on hard negatives introduces a critical vulnerability: the problem of ‘false negatives’. Since negatives are typically sampled randomly from the dataset, some may inadvertently belong to the same semantic class as the anchor. Pushing these false negatives away from the anchor is detrimental, as it forces the model to learn spurious features to separate semantically identical concepts, leading to a fragmented and less generalizable representation space. This issue is exacerbated in datasets with high class overlap, such as fine-grained classification benchmarks. Existing approaches to mitigate this problem have notable limitations. Debaised Contrastive Learning (DCL) Chuang et al. (2020) proposes a correction to the contrastive loss based on an estimated global prior of the false negative rate. This approach is static and assumes a uniform distribution of false negatives, failing to adapt to the local structure of the embedding space or the specific dynamics of an anchor-negative pair. Other methods, such as that proposed by Ding et al. (2020), use the variance of similarity scores over time as a proxy for sample reliability. The core issue with this heuristic is its ambiguity: a true hard negative stably situated on the class boundary can exhibit low score variance, making it indistinguishable from a false negative whose representation is collapsing onto the anchor’s. A more direct signal is needed. We argue that the

temporal evolution, or trajectory, of the similarity score between an anchor and a negative provides a far more direct and robust signal for identifying false negatives. As a model learns, the representation of a false negative should naturally converge towards its anchor, resulting in a similarity score with a consistently positive gradient. In contrast, a true hard negative will either be successfully repelled or remain at a stable, high-similarity boundary. Our core contribution is a method that captures this trajectory signal in a highly efficient, state-free manner. We introduce Momentum-based Trajectory Debiasing (MTS), a novel mechanism that integrates seamlessly into momentum-based contrastive frameworks. MTS leverages the inherent temporal lag between the fast-updating online encoder and the slow-moving momentum encoder. By comparing the similarity of a negative to the anchor’s online embedding versus its momentum embedding, we derive a proxy for the instantaneous gradient of the similarity score. This ‘Momentum-based Trajectory Score’ (MTS) directly measures whether a negative is being pulled closer, allowing for a dynamic, instance-specific debiasing of the contrastive loss. Our contributions are as follows:

1. We propose a novel, state-free method to approximate the learning trajectory of anchor-negative similarities using the intrinsic temporal gap in momentum-based contrastive learning, avoiding the need for complex state storage.
2. We introduce the Momentum-based Trajectory Score (MTS), an unambiguous, gradient-based signal that is more direct and effective at identifying false negatives than heuristics like score variance.
3. We develop a dynamically debiased contrastive loss that uses the MTS to perform instance-specific correction, significantly improving representation quality, as demonstrated by our experiments.

We validate our approach on the CIFAR-100 benchmark, which serves as a challenging fine-grained classification task. Our results show that MTS substantially outperforms strong baselines, including standard MoCo, DCL, and a score-variance-based method, confirming the efficacy of our trajectory-based debiasing strategy.

2 RELATED WORK

Our work is situated within the broader context of self-supervised contrastive learning and, more specifically, the challenge of negative sampling. The quality of negative samples is paramount for learning effective representations. Initial works in contrastive learning relied on random sampling for negatives. However, subsequent research quickly identified the importance of hard negatives for accelerating and improving training. Robinson et al. (2020) demonstrated that sampling negatives that are hard to distinguish from the anchor leads to representations that better separate semantic classes. Similarly, Kalantidis et al. (2020) explored mixing strategies to synthesize harder negatives. While these methods improve performance, they inherently increase the probability of selecting false negatives—samples that are hard precisely because they share the same underlying semantics as the anchor. Our work does not propose a new sampling strategy but rather a correction mechanism to handle the inevitable false negatives that arise from any hard-negative sampling approach. Several works have directly addressed the false negative problem. The most related is Debiased Contrastive Learning (DCL) by Chuang et al. (2020). DCL modifies the InfoNCE loss by estimating a global probability of a sample being a false negative, assuming a uniform class distribution. It then uses this static prior to correct the denominator of the softmax loss. In contrast, our MTS approach is dynamic and instance-specific. Instead of relying on a global, static prior, we assess the likelihood of a sample being a false negative based on its unique learning trajectory relative to a specific anchor. This allows our method to adapt to the local data manifold and the model’s current learning state, which is particularly crucial in fine-grained domains where the false negative distribution is highly non-uniform. Another line of work attempts to identify unreliable negatives based on their training dynamics. Ding et al. (2020) (SRNS) observe that false negatives tend to have stable, high similarity scores over training iterations. They propose using the variance of prediction scores as a signal, favoring high-variance samples as they are more likely to be true negatives. While related in spirit, this variance-based signal is fundamentally ambiguous. A true hard negative that consistently lies on the decision boundary may also exhibit low score variance, yet it is a highly valuable sample for learning. Our MTS signal is more direct; by approximating the gradient of the similarity score, we specifically target samples

whose representations are collapsing towards the anchor, a hallmark of false negatives, while correctly identifying stable hard boundary samples as true negatives. In the context of dense retrieval, ANCE Xiong et al. (2020) tackles negative sampling by building an Approximate Nearest Neighbor index over the corpus to retrieve more realistic and challenging hard negatives. While effective for its domain, this approach focuses on improving the quality of the negative set rather than correcting for false negatives that may still be present within it. Our method is complementary and can be applied on top of such advanced sampling strategies to further refine the learning signal.

3 BACKGROUND

To understand our method, we first review the foundational concepts of contrastive self-supervised learning, with a focus on the Momentum Contrast (MoCo) framework upon which our work is built. Contrastive learning aims to learn an embedding function that maps semantically similar data points to nearby locations in a representation space and dissimilar points far apart. The dominant objective function for this task is InfoNCE. Given an anchor sample x_q , a semantically similar (‘positive’) sample x_{k+} , and a set of N semantically dissimilar (‘negative’) samples $\{x_{k_i}\}$, an encoder network f maps these to vector representations $q = f(x_q)$, $k+ = f(x_{k+})$, and $\{k_i = f(x_{k_i})\}$. The InfoNCE loss is formulated as a categorical cross-entropy loss that aims to classify $k+$ correctly from the set of all negatives:

$$\mathcal{L}_q = -\log \left(\frac{\exp(\text{sim}(q, k+)/\tau)}{\exp(\text{sim}(q, k+)/\tau) + \sum_i \exp(\text{sim}(q, k_i)/\tau)} \right)$$

Here, $\text{sim}(u, v)$ is the cosine similarity $u^T v / (\|u\| \|v\|)$, and τ is a temperature hyperparameter that controls the sharpness of the distribution. The Momentum Contrast (MoCo) framework Kalantidis et al. (2020) provides an effective and scalable way to implement this objective. Instead of relying on negatives from the current mini-batch, MoCo maintains a large queue of negative representations from past mini-batches. This decouples the batch size from the number of negatives, allowing for a much larger and more consistent negative set. MoCo employs two encoders: an online encoder f_q that processes the anchor x_q , and a momentum encoder f_k that processes the positive and negative keys. The parameters of the momentum encoder, θ_k , are not updated by backpropagation. Instead, they are updated as a moving average of the online encoder’s parameters, θ_q :

$$\theta_k \leftarrow m \cdot \theta_k + (1 - m) \cdot \theta_q$$

where m is a momentum coefficient (e.g., 0.999). This slow-moving momentum encoder provides stable and consistent representations for the keys in the queue, which is critical for effective learning.

3.1 PROBLEM SETTING AND NOTATION

Our work operates within the MoCo framework. Let x_q be an anchor image. We generate two augmented views, v_1 and v_2 . The online query is $q_{\text{online}} = f_q(v_1)$, and the positive key is $k_{\text{pos}} = f_k(v_2)$. The negative keys $\{k_i\}$ are stored in a memory queue of size K , where each k_i is an embedding of a past image computed by f_k . The core problem, known as the ‘false negative’ problem, arises because the unsupervised sampling process does not guarantee that x_q and the source image of a negative key k_i belong to different semantic classes. When they belong to the same class, k_i is a false negative. The standard InfoNCE loss will incorrectly apply a repulsive force between q_{online} and k_i , harming the semantic structure of the learned embedding space. Our goal is to identify these false negatives on-the-fly and mitigate their negative impact on the learning objective.

4 METHOD

We introduce Momentum-based Trajectory Debiasing (MTS), a mechanism that dynamically identifies and down-weights the influence of potential false negatives by inspecting their learning trajectory. Our method is designed to be state-free and computationally lightweight, integrating seamlessly into momentum-based frameworks like MoCo.

4.1 MOTIVATION AND CORE IDEA

The fundamental premise of our work is that the learning dynamics of a false negative differ predictably from those of a true negative. As the model learns, it should naturally pull representations of same-class instances closer together. Therefore, for a given anchor, the similarity to a false negative should, on average, increase over the course of training. In contrast, a true negative should be pushed away, causing its similarity to decrease or stabilize at a low value. A true *hard* negative might maintain a high similarity but should not exhibit a consistent positive gradient. We propose to capture this temporal gradient as a direct signal for identifying false negatives.

4.2 DUAL-ENCODER SIGNAL FOR TRAJECTORY ESTIMATION

Instead of explicitly tracking similarity scores over multiple training steps, which would require significant memory and state management, we leverage the intrinsic temporal information embedded within the MoCo architecture. The online encoder f_q represents the most current state of the model, while the momentum encoder f_k represents a slowly evolving historical average of the model’s state. This inherent temporal gap provides a natural mechanism for approximating the recent trajectory of a representation. For a given query image x_q , we compute two distinct query representations:

1. The **online query** $q_{\text{online}} = f_q(x_q)$, which reflects the model’s current state.
2. The **momentum query** $q_{\text{momentum}} = f_k(x_q)$, which reflects a recent historical state.

4.3 MOMENTUM-BASED TRAJECTORY SCORE (MTS)

For a given q_{online} , we first identify its top- K_{hard} hardest negatives $\{k_i\}$ from the memory queue based on their current similarity scores. For each of these hard negatives, we compute two similarity values: $s_{\text{current}} = \text{sim}(q_{\text{online}}, k_i)$: The similarity from the perspective of the current model. $s_{\text{historical}} = \text{sim}(q_{\text{momentum}}, k_i)$: The similarity from the perspective of the recent past model. The difference between these scores serves as a first-order approximation of the similarity gradient. We define the Momentum-based Trajectory Score (MTS) as the positive part of this difference:

$$\text{MTS}(q, k_i) = \text{ReLU}(s_{\text{current}} - s_{\text{historical}})$$

A positive MTS value indicates that the online encoder has moved the query representation closer to the negative key k_i compared to where the momentum encoder had placed it. This provides a strong, instance-specific signal that the model is learning to collapse this anchor-negative pair, a characteristic behavior for a false negative.

4.4 DYNAMICALLY DEBIASED CONTRASTIVE LOSS

We integrate the MTS directly into the InfoNCE loss to dynamically correct for suspected false negatives. For each of the top- K_{hard} hard negatives, we adjust its similarity score before it enters the softmax calculation. The adjusted similarity is:

$$\text{sim}_{\text{adjusted}}(q, k_i) = s_{\text{current}} - \lambda \cdot \text{MTS}(q, k_i)$$

Here, λ is a hyperparameter that controls the strength of the debiasing penalty. This adjustment effectively reduces the repulsive force applied to negatives that are identified as potential false positives. The repulsion is attenuated proportionally to the strength of the trajectory signal. For negatives with an MTS of zero (i.e., those being pushed away or remaining stable), the loss remains unchanged, allowing the model to continue learning from true hard negatives. The complete loss calculation then uses $\text{sim}_{\text{adjusted}}$ for the debiased hard negatives and s_{current} for all other negatives. This entire process adds only a single extra forward pass of the query batch through the momentum encoder, making its computational overhead negligible.

5 EXPERIMENTAL SETUP

We conducted experiments to validate the effectiveness of our Momentum-Based Trajectory Debiasing (MTS) method against strong contrastive learning baselines. The experimental design focuses on a fine-grained classification benchmark where the false negative problem is particularly pronounced.

5.1 DATASET AND PRE-TRAINING

We used the CIFAR-100 dataset for both self-supervised pre-training and downstream evaluation. CIFAR-100 consists of 60,000 32x32 color images in 100 classes, grouped into 20 superclasses. This hierarchical class structure ensures a high degree of semantic overlap, creating a challenging environment rich with potential false negatives. For pre-training, we used the 50,000 training images and discarded all labels.

5.2 BACKBONE ARCHITECTURE

A ResNet-18 architecture, adapted for the 32x32 input resolution of CIFAR-100, was used as the backbone encoder for all self-supervised methods. This ensures a fair comparison of the learning objectives themselves.

5.3 METHODS COMPARED

We compared our proposed method, MoCo-MTS, against several baselines:

- **MoCo:** The standard Momentum Contrast v2 framework, serving as our primary baseline.
- **DCL:** We implemented the Debaised Contrastive Learning Chuang et al. (2020) loss on top of the MoCo framework. Following their work, we used a false negative prior of $\tau_{\text{plus}} = 0.1$ for CIFAR-100.
- **SRNS:** A simplified, batch-local implementation of a score-variance-based debiasing method inspired by Ding et al. Ding et al. (2020). This baseline filters negatives based on their score variance as a proxy for reliability.
- **Supervised:** As a practical upper bound, we evaluated a ResNet-18 model pre-trained on ImageNet with full supervision.

5.4 SELF-SUPERVISED PRE-TRAINING CONFIGURATION

All self-supervised models were pre-trained for 100 epochs using the SGD optimizer with a momentum of 0.9 and weight decay of 10^{-4} . The learning rate was initialized to 0.03 and followed a cosine annealing schedule. We used a batch size of 256. The data augmentation pipeline was identical to the one used in MoCo v2, including random resized crops, color jitter, grayscale, and Gaussian blur. For the MoCo framework specifics, we set the temperature $\tau = 0.2$, momentum coefficient $m = 0.999$, and a queue size of 16384. For our method, MoCo-MTS, the debiasing strength was set to $\lambda = 2.0$, and we applied the correction to the top $K_{\text{hard}} = 128$ hardest negatives.

5.5 DOWNSTREAM EVALUATION PROTOCOL

To evaluate the quality of the learned representations, we followed the standard linear probing protocol. After the 100-epoch self-supervised pre-training phase, the weights of the ResNet-18 encoder were frozen. A new linear classifier was trained on top of the frozen features using the labeled CIFAR-100 training set. This classifier was trained for 100 epochs using the Adam optimizer with a learning rate of 10^{-3} . We used simple augmentations (random resized crop and horizontal flip) for training the linear layer.

5.6 EVALUATION METRIC

The primary metric for comparison is the Top-1 classification accuracy on the CIFAR-100 test set. This metric directly reflects the linear separability and semantic quality of the learned representations.

6 RESULTS

We evaluated our proposed Momentum-based Trajectory Debiasing (MTS) method by comparing its downstream classification performance against several strong baselines. The results, summarized in

the table below, demonstrate the clear and significant advantage of our approach on the fine-grained CIFAR-100 benchmark.

Method	Final Top-1 Accuracy (%)
MTS (Ours)	68.21
DCL	65.83
SRNS	65.52
MoCo	64.55
Supervised	72.14

6.1 PERFORMANCE ANALYSIS

Our MoCo-MTS method achieved a Top-1 accuracy of 68.21%, substantially outperforming all other self-supervised methods. This represents a gain of 3.66 percentage points over the standard MoCo baseline (64.55%). This significant improvement underscores the severity of the false negative problem and the effectiveness of our trajectory-based solution in mitigating it. When compared to other debiasing techniques, MTS also shows a clear advantage. It surpasses Debaised Contrastive Learning (DCL) Chuang et al. (2020), the next best method, by 2.38 percentage points. This result supports our hypothesis that a dynamic, instance-specific signal derived from learning trajectories is more effective than a static, global prior for correcting false negatives, especially in datasets with complex semantic structures like CIFAR-100. Similarly, MTS outperforms the score-variance-based SRNS baseline Ding et al. (2020) by 2.69 points, confirming that our direct gradient approximation is a more reliable signal than the ambiguous score variance heuristic. The performance of the supervised ImageNet pre-trained model (72.14%) serves as a practical upper bound. While it remains the top performer, our MTS method successfully closes a large portion of the gap between unsupervised in-domain pre-training and supervised pre-training on a massive external dataset. This highlights the ability of our debiasing strategy to help learn high-quality, semantically robust representations without access to any labels.

6.2 ABLATION AND FAIRNESS CONSIDERATIONS

The key hyperparameters in our method are the penalty strength λ and the number of hard negatives K_{hard} to debias. The reported results use $\lambda = 2.0$ and $K_{\text{hard}} = 128$, which were found to be effective in preliminary experiments. An ablation study where $\lambda = 0$ would recover the MoCo baseline, demonstrating that the entire performance gain is attributable to our trajectory-based correction mechanism. While our method does not introduce any specific fairness-related biases, it is important to note that, like all self-supervised methods, it is susceptible to inheriting and potentially amplifying biases present in the pre-training data. Care must be taken when applying these models in sensitive domains.

6.3 LIMITATIONS

A limitation of our method is the introduction of the λ and K_{hard} hyperparameters, which may require tuning for optimal performance on different datasets or with different model architectures. Furthermore, the efficacy of our method relies on the momentum encoder providing a faithful-yet-lagging representation of the online encoder. This assumption is generally valid in MoCo-style training, but its dynamics could be affected by the choice of the momentum coefficient m .

7 CONCLUSION

In this paper, we addressed the critical problem of false negatives in contrastive self-supervised learning. We argued that existing solutions are insufficient because they rely on static priors or ambiguous heuristics that fail to capture the instance-specific dynamics of the learning process. We proposed Momentum-based Trajectory Debiasing (MTS), a novel, state-free, and computationally efficient method that rectifies this issue. Our core contribution is the insight that the temporal trajectory of similarity scores provides a direct and powerful signal for identifying false negatives. MTS operationalizes this insight by leveraging the inherent temporal gap between the online and momentum encoders in a MoCo-like framework to approximate this trajectory without any explicit state tracking.

By dynamically reducing the repulsive forces on negatives that show signs of representation collapse, MTS encourages the formation of a more semantically coherent embedding space. Our experimental results on the CIFAR-100 fine-grained benchmark validate the effectiveness of our approach. MTS significantly outperformed standard MoCo and prominent debiasing methods like DCL and SRNS, demonstrating its superior ability to learn high-quality representations in the presence of high semantic ambiguity. For future work, the MTS signal itself could be explored as an online diagnostic tool to measure semantic ambiguity within datasets during pre-training. Furthermore, extending the principle of trajectory-based debiasing to other domains, such as natural language processing and reinforcement learning, presents a promising avenue for research.

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REFERENCES

- Ching-Yao Chuang, Joshua Robinson, Lin Yen-Chen, Antonio Torralba, and Stefanie Jegelka. Debaised contrastive learning. *Advances in Neural Information Processing Systems (2020)*, 2020.
- Jingtao Ding, Yuhao Quan, Quanming Yao, Yong Li, and Depeng Jin. Simplify and robustify negative sampling for implicit collaborative filtering. 2020.
- Yannis Kalantidis, Mert Bulent Sariyildiz, Noe Pion, Philippe Weinzaepfel, and Diane Larlus. Hard negative mixing for contrastive learning. 2020.
- Joshua Robinson, Ching-Yao Chuang, Suvrit Sra, and Stefanie Jegelka. Contrastive learning with hard negative samples. 2020.
- Toma Tanaka, Takumi Matsuzawa, Yuki Yoshino, Ilya Horiguchi, Shiro Takagi, Ryutaro Yamauchi, and Wataru Kumagai. AIRAS, 2025. URL <https://github.com/airas-org/airas>.
- Lee Xiong, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul Bennett, Junaid Ahmed, and Arnold Overwijk. Approximate nearest neighbor negative contrastive learning for dense text retrieval. 2020.